Madeleine S. Yuh

(847)-636-1579 | myuh@purdue.edu | www.linkedin.com/in/madeleine-yuh

EDUCATION:

Ph.D., Mechanical Engineering, Purdue University – West Lafayette	**Expected Completion: Dec 2025**
Area: Systems, Measurements, and Control, **Advisor**: Dr. Neera Jain	**Aug 2020 – Present**
M.S., Mechanical Engineering, Purdue University – West Lafayette	**Completed: Aug 2024**
Area: Systems, Measurements, and Control, **Advisor**: Dr. Neera Jain	**Aug 2020 – Aug 2024**
B.S., Mechanical Engineering, University of Illinois at Chicago	**Completed: May 2020**
Summa Cum Laude, UIC Honors College	**Aug 2016 – May 2020**

RESEARCH INTERESTS:

Deterministic & Probabilistic Dynamic Modeling, Optimal Control, Human-Machine Interaction, Machine Learning, Reinforcement Learning, Cyber-Physical Systems, Intelligent Tutoring Systems, Human Automation Teaming, Rehabilitation Robotics.

RELEVANT COURSEWORK:

Multidisciplinary Design Optimization, Optimal Control & Estimation, Nonlinear Feedback Control, Digital Signal Processing, Machine Learning, Reinforcement Learning, System Identification, Statistics.

HONORS & AWARDS:

1. Frederick N. Andrews Fellowship, School of Mechanical Engineering at Purdue University, 2020–2024
2. Summer Undergraduate Research Fellowship (SURF) graduate mentor award, Purdue University, 2022
3. 2022 CPHS Fellows, 4th IFAC Workshop on Cyber-Physical Human Systems, 2022
4. ASPIRE 2024 Alphonse Chapanis student paper award, 2024
5. HFES CEDM technical group best student paper award finalist, 2024
6. Hommert Engineering Excellence Fellowship, 2024
7. Ward A. Lambert Graduate Teaching Fellowship, 2024
8. ICON Student Research Conference 2025 Best Presentation Award, 2025
9. 2025 NSF CPS Rising Star, 2025
10. Purdue OGSPS Mentoring Award for Graduate Students College of Engineering nominee, 2025
11. MECC 2025 DSCD Rising Star, 2025
12. IEEE RAS Travel Award to attend IEEE ROMAN 2025, 2025
13. MIT Rising Stars in Mechanical Engineering, 2025
14. Summer Undergraduate Research Fellowship (SURF) graduate mentor award, Purdue University, 2025

PUBLICATIONS:

Peer Reviewed Journal Articles

- K. J. Williams, **M. S. Yuh**, and N. Jain, "A Computational Model of Coupled Human Trust and Self-Confidence Dynamics", *ACM Transactions on Human-Robot Interaction,* vol. 12, no. 3, pp.1-29, June 2023.
- **M. S. Yuh**, K. R. Ortiz, K. Sommer-Kohrt, M. Oishi and N. Jain, “Classification of Human Learning Stages via Kernel Distribution Embeddings,” *IEEE Open Journal of Control Systems*, pp. 1-16, January 2024.

Peer Reviewed Conference Papers

- **M. S. Yuh**, S. Byeon, I. Hwang, and N. Jain, “A Heuristic Strategy for Cognitive State-based Feedback Control to Accelerate Human Learning,” in *IFAC-PapersOnLine*, vol. 55, no. 41, pp. 107-112, 2022.
- **M. S. Yuh**, E. Rabb, A. Thorpe, and N. Jain, “Using Reward Shaping to Train Cognitive-based Control Policies for Intelligent Tutoring Systems”, in *2024 American Control Conference (ACC*), Toronto, ON, 2024.
- **M. S. Yuh** and N. Jain, “Designing Cognitively-Aware Psychomotor Intelligent Tutoring Systems via Multi-Objective and Human-Centric Optimization” in *2025 IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, Eindhoven, the Netherlands, 2025.

- M. Wielatz, M. Pandya, **M. S. Yuh**, and N. Jain, "The Role of Dispositional Trust in Adaptive Automation for Trust Calibration" in *2025 IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, Eindhoven, the Netherlands, 2025.

Conference Papers

- **M. S. Yuh** and N. Jain, "Online Self-Confidence Calibration for Improving Learning Outcomes via Intelligent Tutoring Systems," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, p. 10711813241260743, Aug. 2024.
    - ASPIRE 2024 Alphonse Chapanis Best Student Paper Award
    - HFES Cognitive Engineering and Decision Making Technical Group Best Student Paper Award Finalist
- K. Ortiz, J. Hunter, A. Thorpe, **M. S. Yuh**, T. Reid, N. Jain and M. Oishi, "Assessing the Relationship Between Learning Stages and Prefrontal Cortex Activation in a Psychomotor Task," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, p. 10711813241260675, Aug. 2024
- S. Byeon, **M. S. Yuh**, J. Choi, N. Jain, and I. Hwang, "Workload Classification for Function Allocations in Human-Autonomy Teaming Using Noninvasive Measurements," in AIAA SCITECH 2025 Forum, Jan. 2025.

Book Chapters:

- N. Jain, T. Reid, K. Akash, **M. S. Yuh**, and J. Hunter, "Enabling Human-Aware Autonomy Through Cognitive Modeling and Feedback Control," in Cyber-Physical-Human Systems: Fundamentals and Applications. A. M. Annaswamy, P. P. Khargonekar, F. Lamnabhi-Lagarrigue, and S. K. Spurgeon, Eds. Hoboken, NJ, USA; John Wiley & Sons, Inc. 2023, ch. 5, pp 95-124.

RESEARCH AND TEACHING EXPERIENCE:

Purdue University – West Lafayette, Indiana

Ward A. Lambert Graduate Teaching Fellowship — Jan 2025 – Present

- Shadowed a faculty member during the spring 2025 semester teaching a section of Systems, Measurements, and Controls 1 (ME 36500). I am currently instructing my own section in the Fall 2025 semester.

Graduate Research Assistant, — Aug 2020 – Present

- Formulated and validated probabilistic model of human states of trust and self-confidence as a dynamic system when interacting with an autonomous system.
- Designed and experimentally validated control algorithms for achieving calibration of human cognitive states in psychomotor learning contexts.
- Synthesized and implemented cognitive-based control policies using reward shaping in an intelligent tutoring system for a psychomotor task.
- Specialization in user studies and control algorithms to infer human cognitive states using behavioral and psycho-physiological data.
- Identified and characterized learning stages quantitatively for psychomotor learning task as well as developed an online learning stage classifier to provide learning stages as inputs to future cognitive control policies.

Midwest Orthopaedics at Rush – Chicago, Illinois

Undergraduate Research Assistant — Sep 2017 – Nov 2018

- Experimentally validated finite element analysis (FEA) simulation data on wear that occurs due to forces experienced in the knee throughout life in metal/poly-ethylene knee implants. Worked with wear testing machines called pin-on-disc and wheel-on-flat to simulate movements of the knee over 1 million cycles.
- Simulated stresses in patellofemoral joint dependent on patella tendon on tibial tubercle position using FEA.
- Used Geomagic Wrap and Truegrid software to smooth and mesh CT scans of bones, cartilage, and tendons.
- Used Abaqus to assemble parts of the Patellofemoral joint to make a model for simulation and FEA.

University of Illinois at Chicago – Chicago, Illinois

ENGR100: Teaching Assistant — Aug 2017 – Dec 2017

- Teaching assistant for UIC's ENGR100: Engineering Success Seminar for mechanical engineering students
- Advised mechanical engineering students about plan of studies, internship resources at UIC and opportunities, and extracurricular activities to gain hands on experience (e.g. the ASME at UIC chapter).

INTERNSHIPS:

Sandia National Laboratories – Albuquerque, New Mexico

AI for Autonomy Intern **June 2024 – Present**

- Designed, simulated, and implemented auctioning algorithms for task-allocation in multi-agent swarm settings.
- Performed system identification to validate laboratory experiments.

Molex LLC – Lisle, Illinois

Manufacturing Integration Engineering Intern **May 2019 – Aug 2019**

- Evaluated product designs, making sure they were designed for manufacturability.
- Determined the best process for manufacturing, focusing on time efficiency, cost, and quality.
- Automated documentation processes for budget and project management using VBA.
- Competed in intern innovation challenge working with four interns on a proposal for an autonomous vehicle parking garage that was presented to global leadership team of Molex.

PROFESSIONAL ACTIVITIES & SERVICE:

Purdue University – West Lafayette, Indiana

ICON Student Research Conference Program Coordinator **Nov 2023 – Apr 2024**

- Project lead on coordinating student research conference committee members with roles and reviewing submitted abstracts for presentation and posters.
- Created website with relevant information pertaining to scheduling, registration, and abstract submission.

ICON Student Research Conference Poster Session Co-chair **Nov 2022 – Apr 2024**

- Organized poster session for the 2023 and 2024 Purdue ICON student research conference for students to offer students the chance to present their research and network with faculty members and keynote speakers and explore new research avenues and interdisciplinary partnerships.

Purdue ME Taskforce on Female Undergraduate Recruitment **Aug 2021 – Jan 2024**

- Graduate student representative of ME Taskforce on Female Undergraduate Recruitment.
- Served as panelist for Women in Mechanical Engineering Symposium for current women in ME student panel.
- Organized poster session for the Women in Mechanical Engineering Symposium to encourage Purdue undergraduates to explore different research topics and consider graduate school after graduation.

SIRI 2023 Graduate Leadership Team **May 2023- Aug 2023**

- Organized 8-week undergraduate research program curriculum to facilitate meaningful research experience aimed towards under-represented students interested in cyber-physical systems.

University of Illinois at Chicago – Chicago, Illinois

ASME (UIC) Student Competitions Officer **May 2017 – May 2020**

- Led robotics team to compete in the Student Professional Design Competition held by ASME every year.
- Trained members to have the skills necessary to design and manufacture a competition ready robot.

MENTORSHIP:

Undergraduate Student Mentor, Purdue University **Aug 2022 – Present**

- Mentoring student working personalizing haptic feedback through reinforcement learning methods based on learner preference.
- Mentored student working on incorporated feedback in an online drone simulation generated through GPT-4 to perturb human self-confidence while learning. This data will be used to extend an existing probabilistic model of human self-confidence behavior and control policy synthesis.
- Mentored student working on further developing an online drone simulation to capture human risk behavior.
- Mentored student running user study on further validating heuristic strategy used to decide when to provide autonomous assistance to users while calibrating self-confidence to skill in learning context. Eye-tracking data used to analyze factors that impact self-confidence and identify learning stages in a drone simulator.

SURF Undergraduate Research Student Mentor, Purdue University **May 2022 – Present**

- **Summer 2025:** Mentoring two students to incorporate haptic feedback into a psychomotor intelligent tutoring system to improve student-tutoring agent communication.
- **Summer 2024:** Mentored student who worked on online updating of probabilistic cognitive state models of human learners as they interact with an intelligent tutoring system for a psychomotor learning task.
- **Summer 2023:** Mentored student on developing an experimental platform and probabilistic model that captures self-confidence and workload perturbations in human-automation teaming contexts.
- **Summer 2022:** Mentored student who incorporated and analyzed eye-tracking data in an in-person user study where users learn to fly a drone in a simulator. Data was used to infer human self-confidence cognitive states.

SIRI Undergraduate Research Student Mentor, Purdue University **May 2021 – Aug 2022**

- **Summer 2022:** Mentored students on developing a probabilistic model on the dynamics of human self-confidence cognitive states when interacting with autonomous assistance in drone landing simulator module.
- **Summer 2021:** Mentored students on designing an online based drone simulation experimental platform designed to collect human cognitive and behavioral data when interaction with automation through autonomous assistance.